# Evaluation runs: **reset** (s5-reset)

generated 2026-11-03 from brain-data b2ae60f86 2026-11-03 17:57:09 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): reset: expected violation; ctl: expected violation (the matched control is a violating one); twin: expected conforms. Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{spec}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 1,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 1,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 1,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 2,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 2,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 2,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 3,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 3,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 3,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 4,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 4,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 4,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 5,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 5,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 5,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 6,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 6,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 6,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 7,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 7,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 7,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 8,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 8,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 8,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 9,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 9,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 9,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 10,1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 10,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 10,4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 10,5 | 120 | 120 | conforms | 0.1 | 2 | 1 h, 250 GB | 1 | 1 | brain01, 2026-11-02 |
| 10,6 | 720 | 720 | conforms | 0.6 | 2 | 1 h, 250 GB | 1 | 1 | brain01, 2026-11-02 |
| 10,7 | 5040 | 5040 | conforms | 4.3 | 2 | 1 h, 250 GB | 1 | 1 | brain01, 2026-11-02 |
| 10,8 | 40320 | 40320 | conforms | 36.3 | 10 | 1 h, 250 GB | 1 | 1 | brain01, 2026-11-02 |
| 10,9 | 362880 | 362880 | conforms | 336.4 | 10 | 1 h, 250 GB | 1 | 1 | brain01, 2026-11-02 |
| 10,10 | 3628800 | 3628800 | conforms | 3500.3 | 10 | 1 h, 250 GB | 1 | 1 | brain01, 2026-11-02 |

### stateless: pair **ctl**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 1,1 | 1 | 2 | 1 | violation | 0.0 | 2 | 1 h, 250 GB | 2 | 2 | brain03, 2026-11-02 |
| 1,3 | 6 | 2 | 1 | violation | 0.0 | 4 | 1 h, 250 GB | 2 | 7 | brain03, 2026-11-02 |
| 1,4 | 24 | 2 | 1 | violation | 0.0 | 2 | 1 h, 250 GB | 2 | 25 | brain03, 2026-11-02 |
| 2,1 | 1 | 4 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | 4 | 4 | brain03, 2026-11-02 |
| 2,3 | 6 | 4 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | 4 | 19 | brain03, 2026-11-02 |
| 2,4 | 24 | 4 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | 4 | 73 | brain03, 2026-11-02 |
| 3,1 | 1 | 8 | 7 | violation | 0.0 | 2 | 1 h, 250 GB | 8 | 8 | brain03, 2026-11-02 |
| 3,3 | 6 | 8 | 7 | violation | 0.0 | 2 | 1 h, 250 GB | 8 | 43 | brain03, 2026-11-02 |
| 3,4 | 24 | 8 | 7 | violation | 0.1 | 2 | 1 h, 250 GB | 8 | 169 | brain03, 2026-11-02 |
| 4,1 | 1 | 16 | 15 | violation | 0.0 | 2 | 1 h, 250 GB | 16 | 16 | brain03, 2026-11-02 |
| 4,3 | 6 | 16 | 15 | violation | 0.0 | 2 | 1 h, 250 GB | 16 | 91 | brain03, 2026-11-02 |
| 4,4 | 24 | 16 | 15 | violation | 0.1 | 2 | 1 h, 250 GB | 16 | 361 | brain03, 2026-11-02 |
| 5,1 | 1 | 32 | 31 | violation | 0.0 | 2 | 1 h, 250 GB | 32 | 32 | brain03, 2026-11-02 |
| 5,3 | 6 | 32 | 31 | violation | 0.1 | 2 | 1 h, 250 GB | 32 | 187 | brain03, 2026-11-02 |
| 5,4 | 24 | 32 | 31 | violation | 0.1 | 2 | 1 h, 250 GB | 32 | 745 | brain03, 2026-11-02 |
| 6,1 | 1 | 64 | 63 | violation | 0.1 | 2 | 1 h, 250 GB | 64 | 64 | brain03, 2026-11-02 |
| 6,3 | 6 | 64 | 63 | violation | 0.1 | 2 | 1 h, 250 GB | 64 | 379 | brain03, 2026-11-02 |
| 6,4 | 24 | 64 | 63 | violation | 0.2 | 2 | 1 h, 250 GB | 64 | 1513 | brain03, 2026-11-02 |
| 7,1 | 1 | 128 | 127 | violation | 0.1 | 4 | 1 h, 250 GB | 128 | 128 | brain03, 2026-11-02 |
| 7,3 | 6 | 128 | 127 | violation | 0.2 | 4 | 1 h, 250 GB | 128 | 763 | brain03, 2026-11-02 |
| 7,4 | 24 | 128 | 127 | violation | 0.5 | 4 | 1 h, 250 GB | 128 | 3049 | brain03, 2026-11-02 |
| 8,1 | 1 | 256 | 255 | violation | 0.3 | 8 | 1 h, 250 GB | 256 | 256 | brain03, 2026-11-02 |
| 8,3 | 6 | 256 | 255 | violation | 0.5 | 8 | 1 h, 250 GB | 256 | 1531 | brain03, 2026-11-02 |
| 8,4 | 24 | 256 | 255 | violation | 1.1 | 6 | 1 h, 250 GB | 256 | 6121 | brain03, 2026-11-02 |
| 9,1 | 1 | 512 | 511 | violation | 0.7 | 16 | 1 h, 250 GB | 512 | 512 | brain03, 2026-11-02 |
| 9,3 | 6 | 512 | 511 | violation | 1.1 | 16 | 1 h, 250 GB | 512 | 3067 | brain03, 2026-11-02 |
| 9,4 | 24 | 512 | 511 | violation | 2.8 | 12 | 1 h, 250 GB | 512 | 12265 | brain03, 2026-11-02 |
| 10,1 | 1 | 1024 | 1023 | violation | 1.8 | 32 | 1 h, 250 GB | 1024 | 1024 | brain03, 2026-11-02 |
| 10,3 | 6 | 1024 | 1023 | violation | 2.8 | 30 | 1 h, 250 GB | 1024 | 6139 | brain03, 2026-11-02 |
| 10,4 | 24 | 1024 | 1023 | violation | 6.4 | 26 | 1 h, 250 GB | 1024 | 24553 | brain03, 2026-11-02 |
| 10,5 | 120 | 1024 | 1023 | violation | 27.4 | 30 | 1 h, 250 GB | 1024 | 122761 | brain01, 2026-11-02 |
| 10,6 | 720 | 1024 | 1023 | violation | 156.1 | 30 | 1 h, 250 GB | 1024 | 736561 | brain01, 2026-11-02 |
| 10,7 | 5040 | 1024 | 1023 | violation | 1101.9 | 24 | 1 h, 250 GB | 1024 | 5155921 | brain01, 2026-11-02 |
| 10,8 | 40320 | 1024 | 1023 | violation | 9207.0 | 26 | 48 h, 250 GB | 1024 | 41247361 | brain01, 2026-11-03 |

### stateless: pair **reset**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 1,1 | 1 | 2 | 1 | violation | 0.0 | 2 | 1 h, 250 GB | 2 | 2 | brain03, 2026-11-02 |
| 1,3 | 6 | 2 | 1 | violation | 0.0 | 2 | 1 h, 250 GB | 2 | 7 | brain03, 2026-11-02 |
| 1,4 | 24 | 2 | 1 | violation | 0.0 | 2 | 1 h, 250 GB | 2 | 25 | brain03, 2026-11-02 |
| 2,1 | 1 | 4 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | 4 | 4 | brain03, 2026-11-02 |
| 2,3 | 6 | 4 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | 4 | 19 | brain03, 2026-11-02 |
| 2,4 | 24 | 4 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | 4 | 73 | brain03, 2026-11-02 |
| 3,1 | 1 | 8 | 7 | violation | 0.0 | 2 | 1 h, 250 GB | 8 | 8 | brain03, 2026-11-02 |
| 3,3 | 6 | 8 | 7 | violation | 0.0 | 2 | 1 h, 250 GB | 8 | 43 | brain03, 2026-11-02 |
| 3,4 | 24 | 8 | 7 | violation | 0.0 | 2 | 1 h, 250 GB | 8 | 169 | brain03, 2026-11-02 |
| 4,1 | 1 | 16 | 15 | violation | 0.0 | 2 | 1 h, 250 GB | 16 | 16 | brain03, 2026-11-02 |
| 4,3 | 6 | 16 | 15 | violation | 0.0 | 2 | 1 h, 250 GB | 16 | 91 | brain03, 2026-11-02 |
| 4,4 | 24 | 16 | 15 | violation | 0.1 | 2 | 1 h, 250 GB | 16 | 361 | brain03, 2026-11-02 |
| 5,1 | 1 | 32 | 31 | violation | 0.0 | 2 | 1 h, 250 GB | 32 | 32 | brain03, 2026-11-02 |
| 5,3 | 6 | 32 | 31 | violation | 0.1 | 2 | 1 h, 250 GB | 32 | 187 | brain03, 2026-11-02 |
| 5,4 | 24 | 32 | 31 | violation | 0.1 | 2 | 1 h, 250 GB | 32 | 745 | brain03, 2026-11-02 |
| 6,1 | 1 | 64 | 63 | violation | 0.1 | 2 | 1 h, 250 GB | 64 | 64 | brain03, 2026-11-02 |
| 6,3 | 6 | 64 | 63 | violation | 0.1 | 2 | 1 h, 250 GB | 64 | 379 | brain03, 2026-11-02 |
| 6,4 | 24 | 64 | 63 | violation | 0.2 | 2 | 1 h, 250 GB | 64 | 1513 | brain03, 2026-11-02 |
| 7,1 | 1 | 128 | 127 | violation | 0.1 | 4 | 1 h, 250 GB | 128 | 128 | brain03, 2026-11-02 |
| 7,3 | 6 | 128 | 127 | violation | 0.3 | 6 | 1 h, 250 GB | 128 | 763 | brain03, 2026-11-02 |
| 7,4 | 24 | 128 | 127 | violation | 0.5 | 4 | 1 h, 250 GB | 128 | 3049 | brain03, 2026-11-02 |
| 8,1 | 1 | 256 | 255 | violation | 0.3 | 10 | 1 h, 250 GB | 256 | 256 | brain03, 2026-11-02 |
| 8,3 | 6 | 256 | 255 | violation | 0.5 | 8 | 1 h, 250 GB | 256 | 1531 | brain03, 2026-11-02 |
| 8,4 | 24 | 256 | 255 | violation | 1.1 | 8 | 1 h, 250 GB | 256 | 6121 | brain03, 2026-11-02 |
| 9,1 | 1 | 512 | 511 | violation | 0.7 | 16 | 1 h, 250 GB | 512 | 512 | brain03, 2026-11-02 |
| 9,3 | 6 | 512 | 511 | violation | 1.0 | 18 | 1 h, 250 GB | 512 | 3067 | brain03, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 9,4 | 24 | 512 | 511 | violation | 2.4 | 16 | 1 h, 250 GB | 512 | 12265 | brain03, 2026-11-02 |
| 10,1 | 1 | 1024 | 1023 | violation | 1.2 | 30 | 1 h, 250 GB | 1024 | 1024 | brain03, 2026-11-02 |
| 10,3 | 6 | 1024 | 1023 | violation | 2.0 | 30 | 1 h, 250 GB | 1024 | 6139 | brain03, 2026-11-02 |
| 10,4 | 24 | 1024 | 1023 | violation | 4.6 | 34 | 1 h, 250 GB | 1024 | 24553 | brain03, 2026-11-02 |
| 10,5 | 120 | 1024 | 1023 | violation | 19.6 | 30 | 1 h, 250 GB | 1024 | 122761 | brain01, 2026-11-02 |
| 10,6 | 720 | 1024 | 1023 | violation | 113.2 | 34 | 1 h, 250 GB | 1024 | 736561 | brain01, 2026-11-02 |
| 10,7 | 5040 | 1024 | 1023 | violation | 801.4 | 30 | 1 h, 250 GB | 1024 | 5155921 | brain01, 2026-11-02 |
| 10,8 | 40320 | 1024 | 1023 | violation | 6572.0 | 30 | 48 h, 250 GB | 1024 | 41247361 | brain01, 2026-11-03 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): twin 10 8 bag, ctl 10 8 bag, reset 10 8 bag, ctl 10 9 bag, reset 10 9 bag, reset 10 10 bag, ctl 10 10 bag, twin 11 1 bag, reset 11 1 bag, ctl 11 1 bag, reset 11 3 bag, ctl 11 3 bag, twin 11 3 bag, twin 11 4 bag, reset 11 4 bag, ctl 11 4 bag, ctl 12 1 bag, reset 12 1 bag, twin 12 1 bag, twin 12 3 bag, ctl 12 3 bag, reset 12 3 bag, ctl 12 4 bag, twin 12 4 bag, … (38 instances in all).

# stateful

## stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 1,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 1,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 1,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 9,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 9,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 9,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,5 | 120 | 120 | conforms | 0.1 | 2 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,6 | 720 | 720 | conforms | 0.3 | 2 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,7 | 5040 | 5040 | conforms | 2.1 | 2 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,8 | 40320 | 40320 | conforms | 17.1 | 10 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,9 | 362880 | 362880 | conforms | 157.8 | 10 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,10 | 3628800 | 3628800 | conforms | 1601.8 | 10 | 1 h, 250 GB | brain01, 2026-11-02 |

## stateful: pair **ctl**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 1,1 | 1 | 2 | 1 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 1,3 | 6 | 2 | 1 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 1,4 | 24 | 2 | 1 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,1 | 1 | 4 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 6 | 4 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 24 | 4 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,1 | 1 | 8 | 7 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 6 | 8 | 7 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 24 | 8 | 7 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,1 | 1 | 16 | 15 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 6 | 16 | 15 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | 24 | 16 | 15 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,1 | 1 | 32 | 31 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3 | 6 | 32 | 31 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,4 | 24 | 32 | 31 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,1 | 1 | 64 | 63 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,3 | 6 | 64 | 63 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,4 | 24 | 64 | 63 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,1 | 1 | 128 | 127 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,3 | 6 | 128 | 127 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,4 | 24 | 128 | 127 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,1 | 1 | 256 | 255 | violation | 0.1 | 8 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,3 | 6 | 256 | 255 | violation | 0.1 | 8 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,4 | 24 | 256 | 255 | violation | 0.1 | 8 | 1 h, 250 GB | brain02, 2026-11-02 |
| 9,1 | 1 | 512 | 511 | violation | 0.2 | 14 | 1 h, 250 GB | brain02, 2026-11-02 |
| 9,3 | 6 | 512 | 511 | violation | 0.3 | 14 | 1 h, 250 GB | brain02, 2026-11-02 |
| 9,4 | 24 | 512 | 511 | violation | 0.3 | 14 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,1 | 1 | 1024 | 1023 | violation | 0.8 | 34 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,3 | 6 | 1024 | 1023 | violation | 0.8 | 36 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,4 | 24 | 1024 | 1023 | violation | 0.8 | 36 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,5 | 120 | 1024 | 1023 | violation | 0.8 | 34 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,6 | 720 | 1024 | 1023 | violation | 0.9 | 34 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,7 | 5040 | 1024 | 1023 | violation | 1.8 | 34 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,8 | 40320 | 1024 | 1023 | violation | 9.0 | 34 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,9 | 362880 | 1024 | 1023 | violation | 77.5 | 34 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,10 | 3628800 | 1024 | 1023 | violation | 791.0 | 34 | 1 h, 250 GB | brain01, 2026-11-02 |

### stateful: pair **reset**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 1,1 | 1 | 2 | 1 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 1,3 | 6 | 2 | 1 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 1,4 | 24 | 2 | 1 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,1 | 1 | 4 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 6 | 4 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 24 | 4 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,1 | 1 | 8 | 7 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 6 | 8 | 7 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 24 | 8 | 7 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,1 | 1 | 16 | 15 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 6 | 16 | 15 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | 24 | 16 | 15 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,1 | 1 | 32 | 31 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3 | 6 | 32 | 31 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,4 | 24 | 32 | 31 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,1 | 1 | 64 | 63 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,3 | 6 | 64 | 63 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,4 | 24 | 64 | 63 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,1 | 1 | 128 | 127 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,3 | 6 | 128 | 127 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,4 | 24 | 128 | 127 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,1 | 1 | 256 | 255 | violation | 0.1 | 8 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,3 | 6 | 256 | 255 | violation | 0.1 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8,4 | 24 | 256 | 255 | violation | 0.1 | 8 | 1 h, 250 GB | brain02, 2026-11-02 |
| 9,1 | 1 | 512 | 511 | violation | 0.2 | 14 | 1 h, 250 GB | brain02, 2026-11-02 |
| 9,3 | 6 | 512 | 511 | violation | 0.2 | 18 | 1 h, 250 GB | brain02, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 9,4 | 24 | 512 | 511 | violation | 0.2 | 16 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,1 | 1 | 1024 | 1023 | violation | 0.5 | 38 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,3 | 6 | 1024 | 1023 | violation | 0.5 | 36 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,4 | 24 | 1024 | 1023 | violation | 0.5 | 36 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10,5 | 120 | 1024 | 1023 | violation | 0.6 | 34 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,6 | 720 | 1024 | 1023 | violation | 0.6 | 36 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,7 | 5040 | 1024 | 1023 | violation | 1.6 | 32 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,8 | 40320 | 1024 | 1023 | violation | 9.0 | 34 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,9 | 362880 | 1024 | 1023 | violation | 79.8 | 32 | 1 h, 250 GB | brain01, 2026-11-02 |
| 10,10 | 3628800 | 1024 | 1023 | violation | 807.5 | 32 | 1 h, 250 GB | brain01, 2026-11-02 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): twin 10 8 bag, ctl 10 8 bag, reset 10 8 bag, twin 11 1 bag, reset 11 1 bag, ctl 11 1 bag, reset 11 3 bag, ctl 11 3 bag, twin 11 3 bag, twin 11 4 bag, reset 11 4 bag, ctl 11 4 bag, ctl 12 1 bag, reset 12 1 bag, twin 12 1 bag, twin 12 3 bag, ctl 12 3 bag, reset 12 3 bag, ctl 12 4 bag, twin 12 4 bag, reset 12 4 bag, reset 13 1 bag, twin 13 1 bag, ctl 13 1 bag, … (34 instances in all).

# offline

## offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1 | 1 | 1 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 1,3 | 6 | 6 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 1,4 | 24 | 24 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,1 | 1 | 1 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,3 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,4 | 24 | 24 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,1 | 1 | 1 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,4 | 24 | 24 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,1 | 1 | 1 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,3 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,4 | 24 | 24 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5,1 | 1 | 1 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5,3 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.0 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5,4 | 24 | 24 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6,1 | 1 | 1 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6,3 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6,4 | 24 | 24 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7,1 | 1 | 1 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7,3 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7,4 | 24 | 24 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,1 | 1 | 1 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,3 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,4 | 24 | 24 | conforms | 0.4 | 4 | 10 min, 16 GB | 0.4 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 9,1 | 1 | 1 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 9,3 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 9,4 | 24 | 24 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 10,1 | 1 | 1 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 10,3 | 6 | 6 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 10,4 | 24 | 24 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 10,5 | 120 | 120 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 10,6 | 720 | 720 | conforms | 0.5 | 4 | 10 min, 16 GB | 0.3 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 10,7 | 5040 | 5040 | conforms | 2.7 | 4 | 10 min, 16 GB | 1.4 | 1.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 10,8 | 40320 | 40320 | conforms | 21.3 | 10 | 10 min, 16 GB | 10.7 | 10.7 | 0.0 | in-memory | brain01, 2026-11-02 |
| 10,9 | 362880 | 362880 | conforms | 195.7 | 9 | 1 h, 250 GB | 98.5 | 97.3 | 0.0 | merge | brain01, 2026-11-03 |
| 10,10 | 3628800 | 3628800 | conforms | 1988.3 | 10 | 1 h, 250 GB | 992.2 | 996.2 | 0.0 | merge | brain01, 2026-11-03 |

## offline: pair **ctl**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1 | 1 | 2 | 1 | violation | 0.2 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 1,3 | 6 | 2 | 1 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 1,4 | 24 | 2 | 1 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,1 | 1 | 4 | 3 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,3 | 6 | 4 | 3 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,4 | 24 | 4 | 3 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,1 | 1 | 8 | 7 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 6 | 8 | 7 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,4 | 24 | 8 | 7 | violation | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,1 | 1 | 16 | 15 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,3 | 6 | 16 | 15 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,4 | 24 | 16 | 15 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5,1 | 1 | 32 | 31 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5,3 | 6 | 32 | 31 | violation | 0.2 | 4 | 10 min, 16 GB | 0.0 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5,4 | 24 | 32 | 31 | violation | 0.4 | 4 | 10 min, 16 GB | 0.2 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6,1 | 1 | 64 | 63 | violation | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6,3 | 6 | 64 | 63 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6,4 | 24 | 64 | 63 | violation | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7,1 | 1 | 128 | 127 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7,3 | 6 | 128 | 127 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7,4 | 24 | 128 | 127 | violation | 0.3 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,1 | 1 | 256 | 255 | violation | 0.2 | 6 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,3 | 6 | 256 | 255 | violation | 0.2 | 6 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,4 | 24 | 256 | 255 | violation | 0.5 | 6 | 10 min, 16 GB | 0.4 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 9,1 | 1 | 512 | 511 | violation | 0.3 | 12 | 10 min, 16 GB | 0.1 | 0.1 | 0.1 | in-memory | brain01, 2026-11-02 |
| 9,3 | 6 | 512 | 511 | violation | 0.5 | 12 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 9,4 | 24 | 512 | 511 | violation | 0.2 | 12 | 10 min, 16 GB | 0.1 | 0.1 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,1 | 1 | 1024 | 1023 | violation | 0.5 | 24 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,3 | 6 | 1024 | 1023 | violation | 0.6 | 24 | 10 min, 16 GB | 0.2 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,4 | 24 | 1024 | 1023 | violation | 0.6 | 24 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,5 | 120 | 1024 | 1023 | violation | 0.6 | 24 | 10 min, 16 GB | 0.1 | 0.4 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,6 | 720 | 1024 | 1023 | violation | 0.8 | 24 | 10 min, 16 GB | 0.3 | 0.4 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,7 | 5040 | 1024 | 1023 | violation | 1.9 | 24 | 10 min, 16 GB | 1.4 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,8 | 40320 | 1024 | 1023 | violation | 11.1 | 9 | 1 h, 250 GB | 10.7 | 0.3 | 0.1 | merge | brain01, 2026-11-03 |

## offline: pair **reset**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1 | 1 | 2 | 1 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 1,3 | 6 | 2 | 1 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 1,4 | 24 | 2 | 1 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,1 | 1 | 4 | 3 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,3 | 6 | 4 | 3 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,4 | 24 | 4 | 3 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,1 | 1 | 8 | 7 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 6 | 8 | 7 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,4 | 24 | 8 | 7 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,1 | 1 | 16 | 15 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,3 | 6 | 16 | 15 | violation | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4,4 | 24 | 16 | 15 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5,1 | 1 | 32 | 31 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5,3 | 6 | 32 | 31 | violation | 0.2 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5,4 | 24 | 32 | 31 | violation | 0.2 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6,1 | 1 | 64 | 63 | violation | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6,3 | 6 | 64 | 63 | violation | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6,4 | 24 | 64 | 63 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7,1 | 1 | 128 | 127 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7,3 | 6 | 128 | 127 | violation | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7,4 | 24 | 128 | 127 | violation | 0.3 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,1 | 1 | 256 | 255 | violation | 0.3 | 6 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,3 | 6 | 256 | 255 | violation | 0.2 | 6 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8,4 | 24 | 256 | 255 | violation | 0.6 | 6 | 10 min, 16 GB | 0.4 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 9,1 | 1 | 512 | 511 | violation | 0.3 | 12 | 10 min, 16 GB | 0.1 | 0.2 | 0.1 | in-memory | brain01, 2026-11-02 |
| 9,3 | 6 | 512 | 511 | violation | 0.5 | 12 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 9,4 | 24 | 512 | 511 | violation | 0.4 | 12 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,1 | 1 | 1024 | 1023 | violation | 0.5 | 24 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,3 | 6 | 1024 | 1023 | violation | 0.7 | 24 | 10 min, 16 GB | 0.2 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,4 | 24 | 1024 | 1023 | violation | 0.6 | 24 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,5 | 120 | 1024 | 1023 | violation | 0.5 | 24 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,6 | 720 | 1024 | 1023 | violation | 0.7 | 24 | 10 min, 16 GB | 0.3 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,7 | 5040 | 1024 | 1023 | violation | 1.8 | 24 | 10 min, 16 GB | 1.4 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 10,8 | 40320 | 1024 | 1023 | violation | 11.5 | 9 | 1 h, 250 GB | 10.7 | 0.3 | 0.5 | merge | brain01, 2026-11-03 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): twin 11 1 bag, reset 11 1 bag, ctl 11 1 bag, reset 11 3 bag, ctl 11 3 bag, twin 11 3 bag, twin 11 4 bag, reset 11 4 bag, ctl 11 4 bag, ctl 12 1 bag, reset 12 1 bag, twin 12 1 bag, twin 12 3 bag, ctl 12 3 bag, reset 12 3 bag, twin 12 4 bag, ctl 12 4 bag, reset 12 4 bag, reset 13 1 bag, twin 13 1 bag, ctl 13 1 bag, twin 13 3 bag, reset 13 3 bag, ctl 13 3 bag, … (27 instances in all).

refused (an input missing at run time): reset 10 9 bag, ctl 10 9 bag, ctl 10 10 bag, reset 10 10 bag, reset 20 11 bag.